Lab 04 Manual

Haleemah Anwary

Seneca College

SRT411NAA: Digital Data Analytics

Kosalan Balarajah

Contents

Setting Up PowerBI (Task 01) 3

Setting Up PowerBI Desktop (Task 02) 3

Connect Sheets and Datastore to Power BI (Task 03) 4

A: 4

B: 6

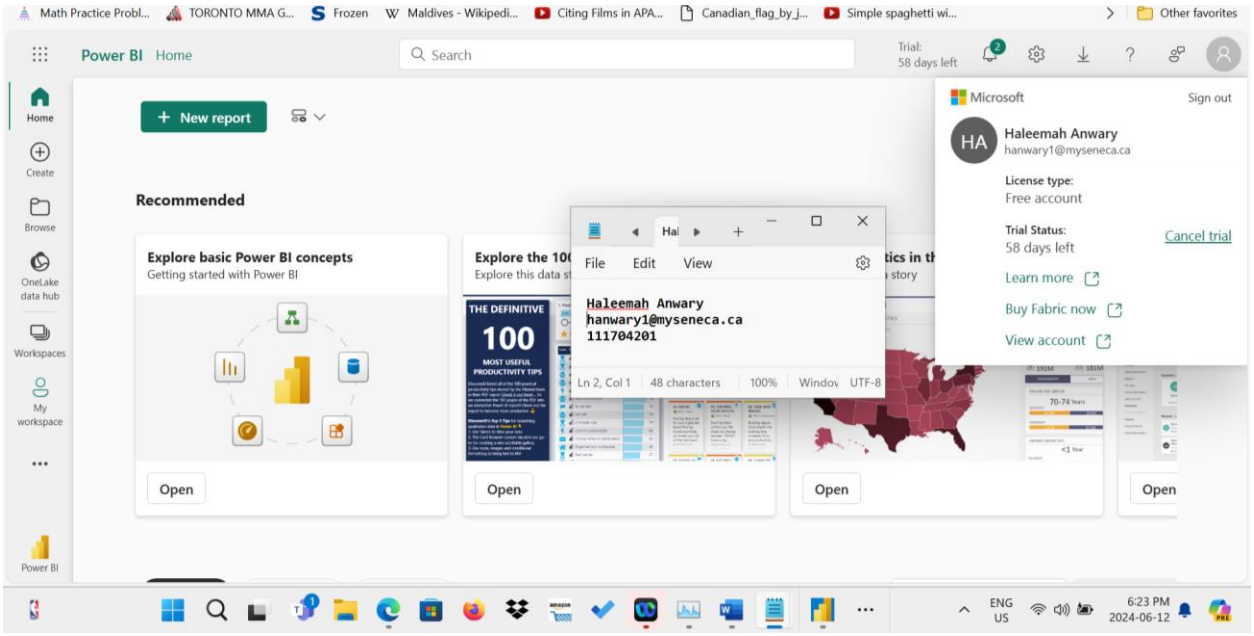
Connect MYSQL and Datastore to Power BI (Task 04) 7

Learning Outcome on Power BI (Task 05) 14

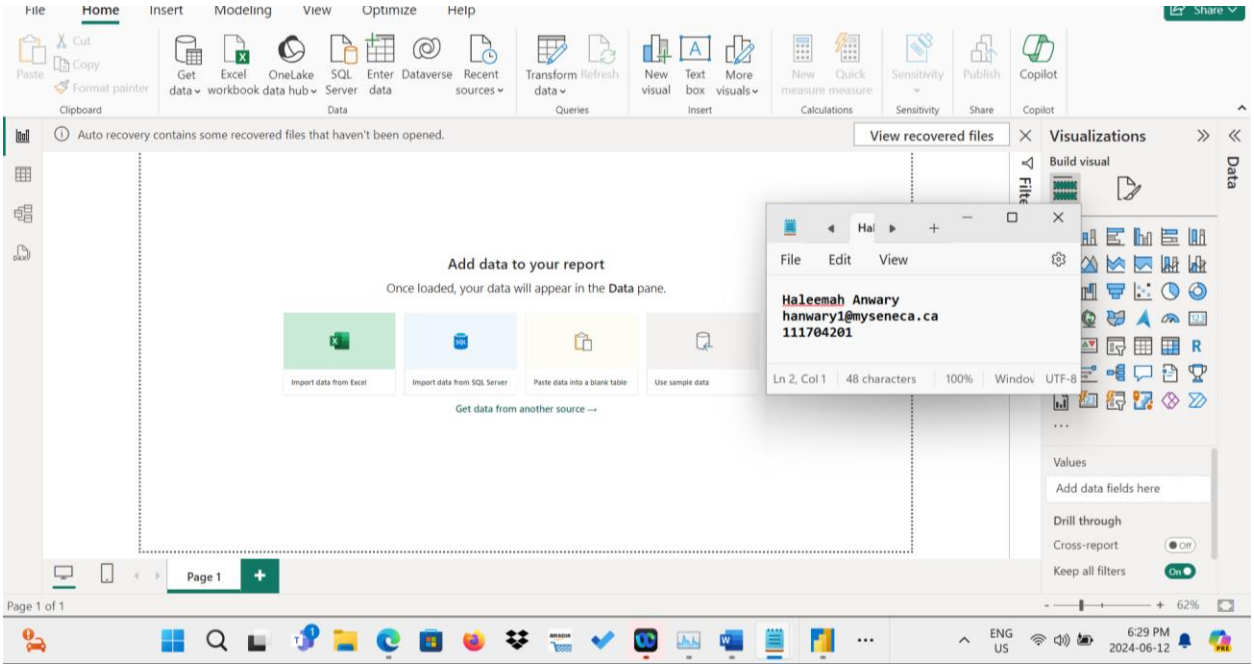
Learning Outcome on Tools and Analytics (Task 06) 15

Learning Outcome In Data Driven Security (Task 07) 15

Setting Up PowerBI (Task 01)



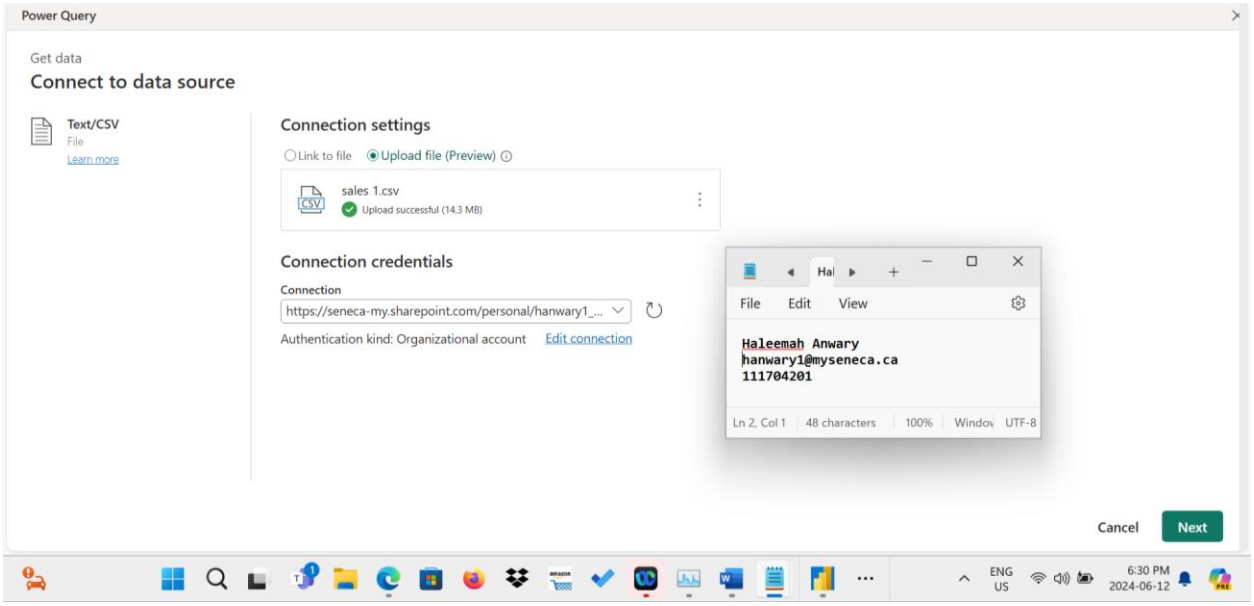
Setting Up PowerBI Desktop (Task 02)



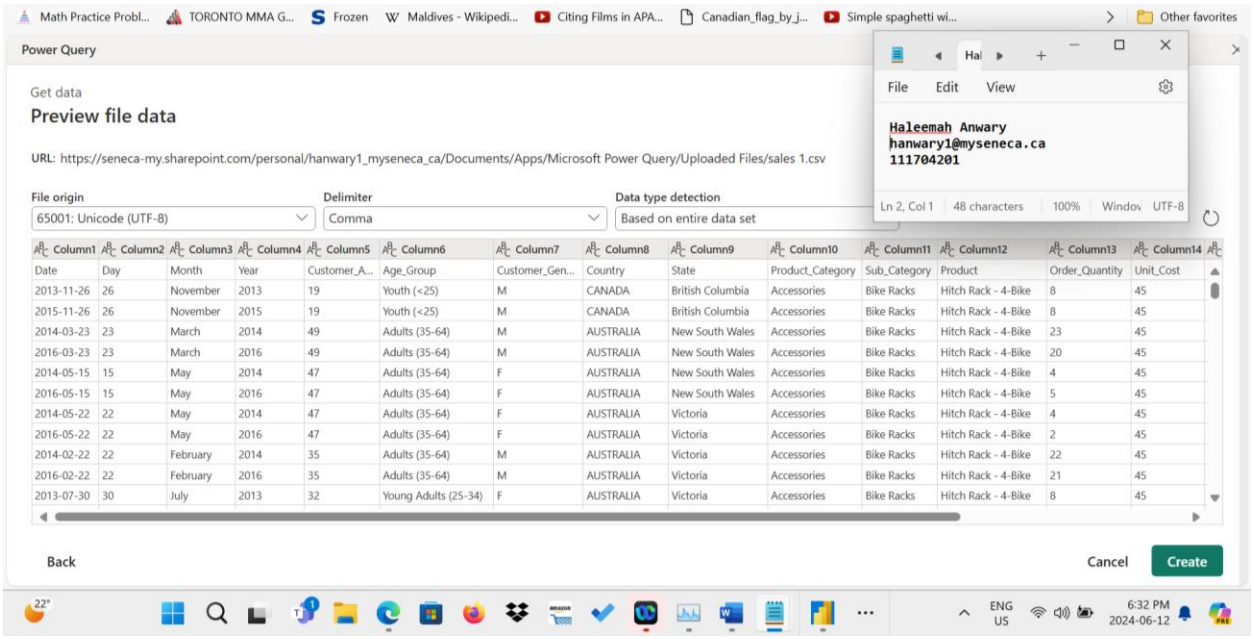
Connect Sheets and Datastore to Power BI (Task 03)

A:

Using Power Query to Connect to the excel sheet



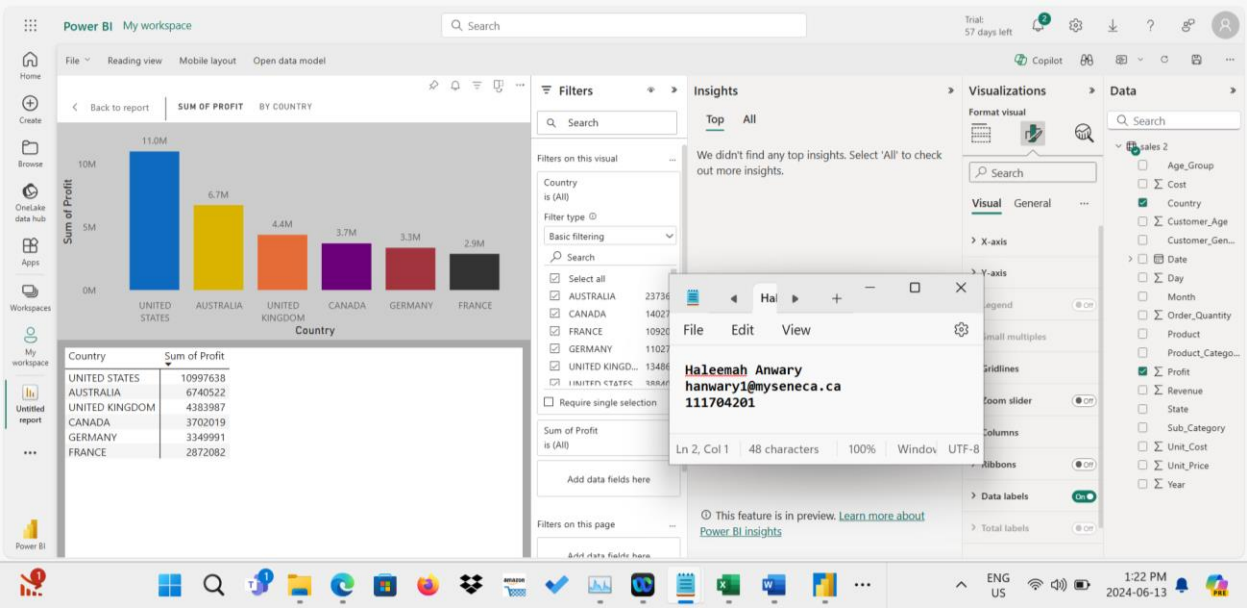
Using the Power Query Editor Wizard create and ingest all the data from Excel



Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8	Column9	Column10	Column11	Column12	Column13	Column14
Date	Day	Month	Year	Customer_A...	Age_Group	Customer_Gen...	Country	State	Product_Category	Sub_Category	Product	Order_Quantity	Unit_Cost
2013-11-26	26	November	2013	19	Youth (<25)	M	CANADA	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	8	45
2015-11-26	26	November	2015	19	Youth (<25)	M	CANADA	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	8	45
2014-03-23	23	March	2014	49	Adults (35-64)	M	AUSTRALIA	New South Wales	Accessories	Bike Racks	Hitch Rack - 4-Bike	23	45
2016-03-23	23	March	2016	49	Adults (35-64)	M	AUSTRALIA	New South Wales	Accessories	Bike Racks	Hitch Rack - 4-Bike	20	45
2014-05-15	15	May	2014	47	Adults (35-64)	F	AUSTRALIA	New South Wales	Accessories	Bike Racks	Hitch Rack - 4-Bike	4	45
2016-05-15	15	May	2016	47	Adults (35-64)	F	AUSTRALIA	New South Wales	Accessories	Bike Racks	Hitch Rack - 4-Bike	5	45
2014-05-22	22	May	2014	47	Adults (35-64)	F	AUSTRALIA	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike	4	45
2016-05-22	22	May	2016	47	Adults (35-64)	F	AUSTRALIA	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike	2	45
2014-02-22	22	February	2014	35	Adults (35-64)	M	AUSTRALIA	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike	22	45
2016-02-22	22	February	2016	35	Adults (35-64)	M	AUSTRALIA	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike	21	45
2013-07-30	30	July	2013	32	Young Adults (25-34)	F	AUSTRALIA	Victoria	Accessories	Bike Racks	Hitch Rack - 4-Bike	8	45

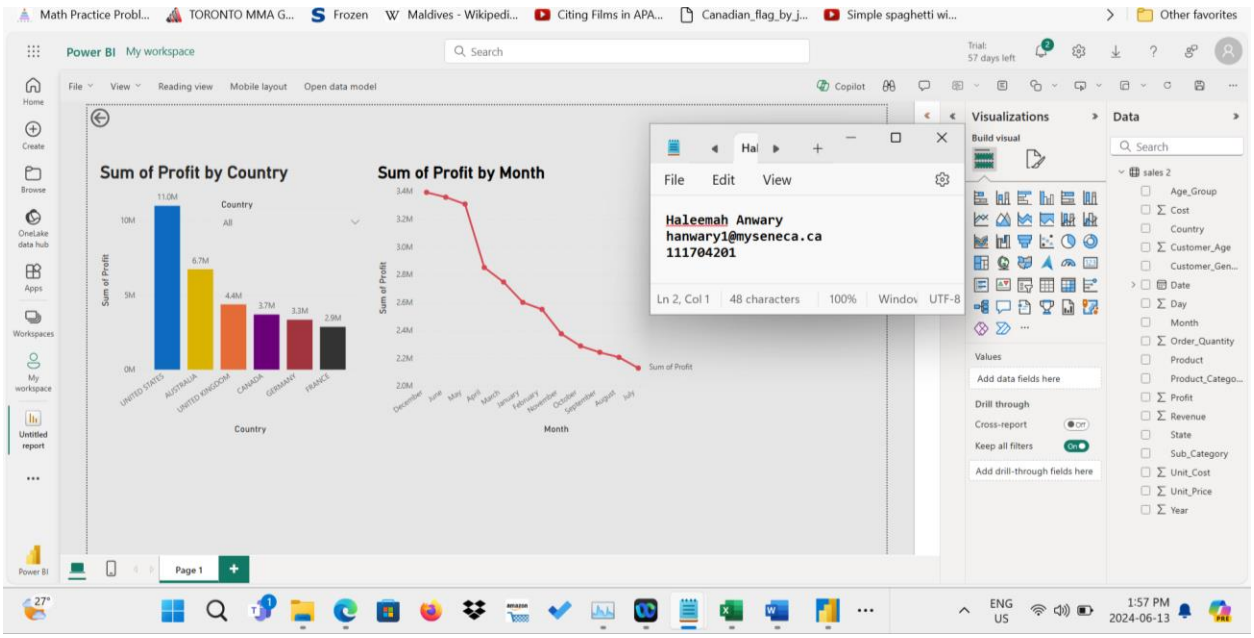
Using the exploration wizard, learn the features of the data.

Explanation: In this example I created the table sum of profit by country and displayed it with the visualization feature using the stacked column chart. I added the labels and changed the background colour, including different colors for each country.



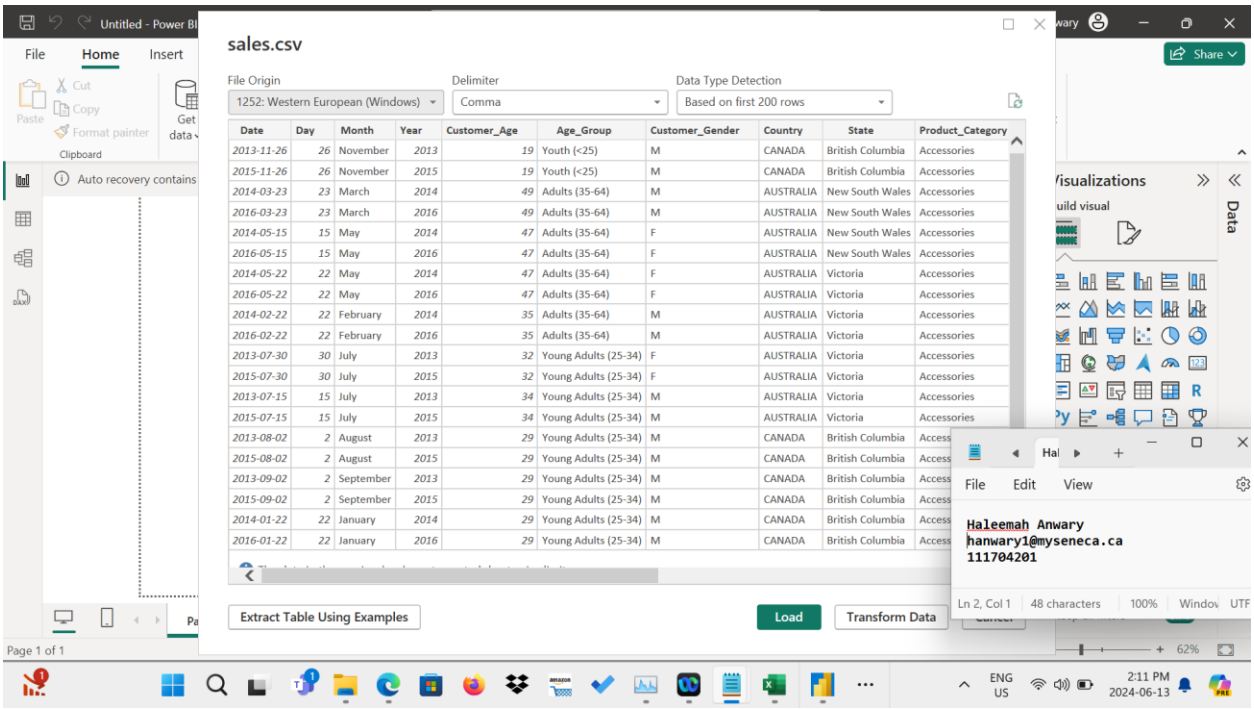
Create a workable report using the Power BI Web Version

In this report, I inserted 2 chart a stacked column chart and a line chart based on the columns, profit, Country and Month. I formatted each one with different colours and adjusted the font and background colors. I also included a slicer here that will interact with all the visuals based on the country column and formatted this to include selection for easier navigation.



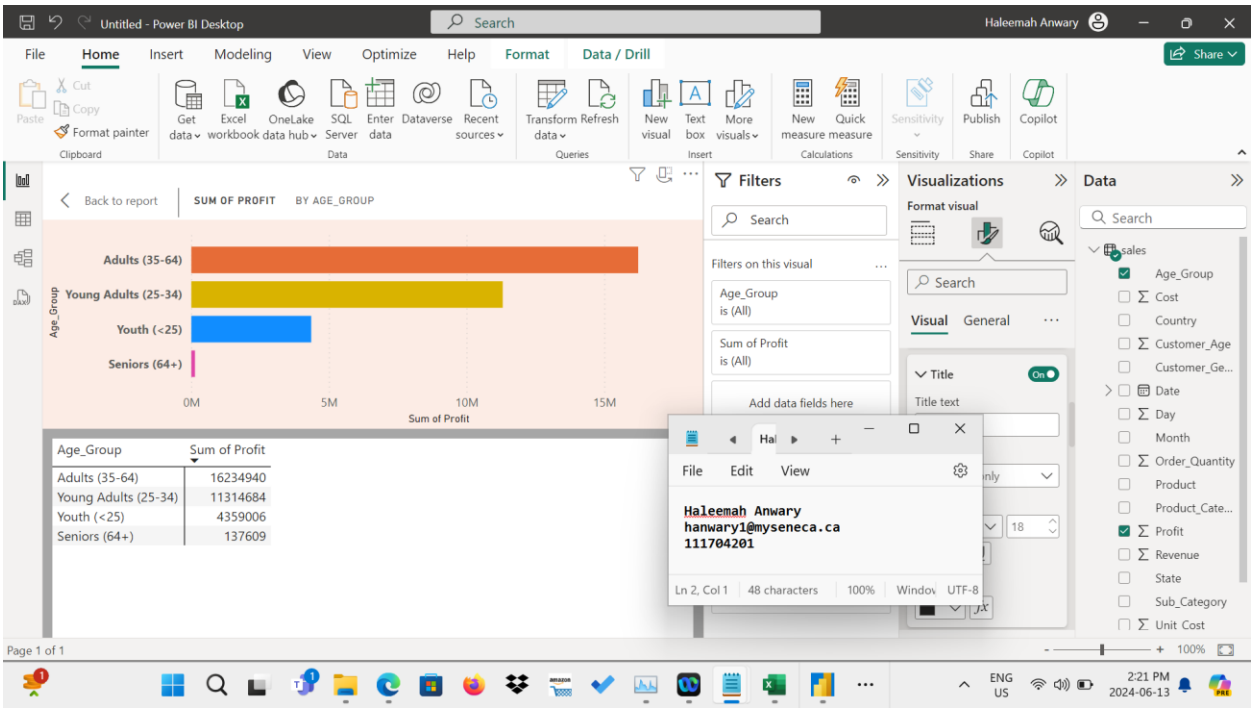
PART B:

Extracting my csv file and loading it into Power BI Desktop



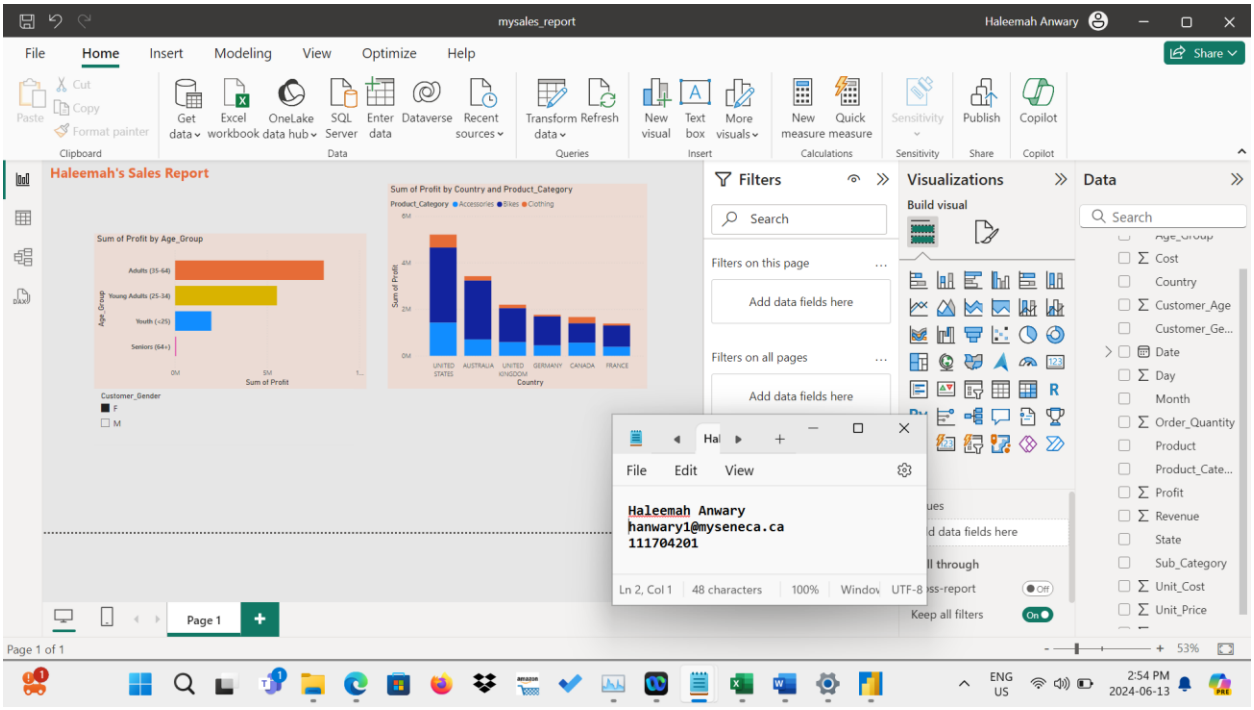
Using the exploration wizard, learn the features of the data

Here, I have sum of profit by age_group to see which age group is generating the most profit. I used a stack bar graph and changed the colors of the bars.



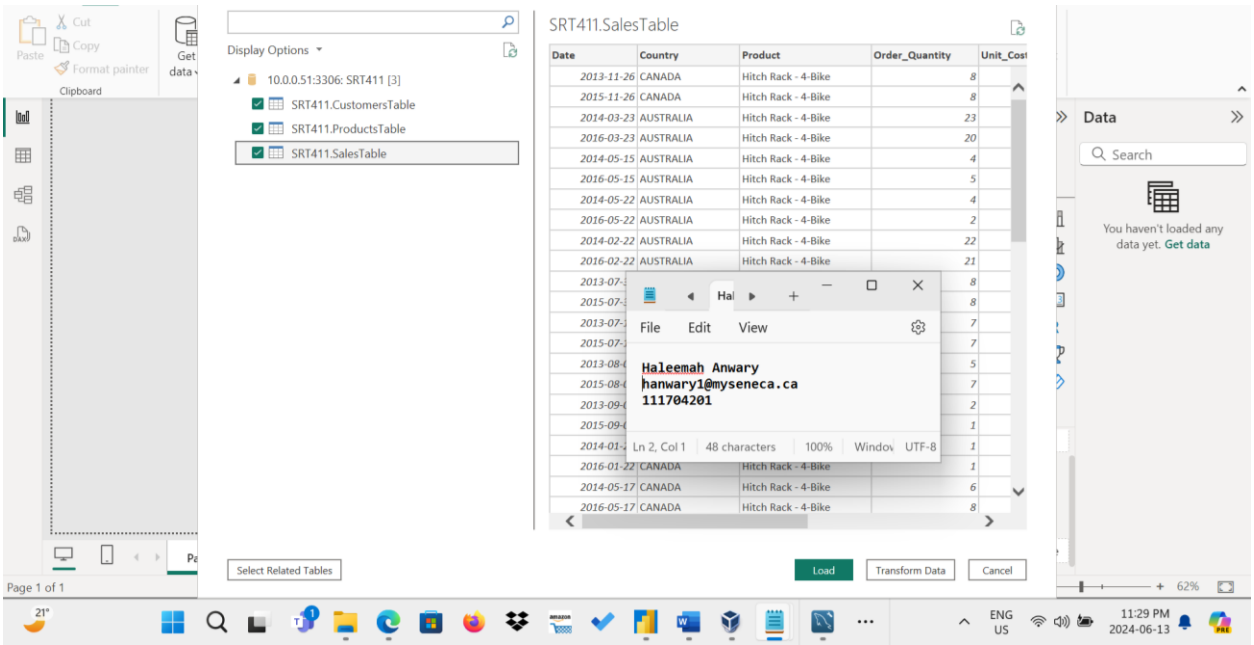
Create a workable report using the Power BI Desktop.

Here I made 2 charts ‘sum of profit by Age_Group’ and another chart showing the sum of profit by country and product_category. I customized the bar colors, titles, and background color to make it easier to read.



Connect MYSQL and Datastore to Power BI (Task 04)

I have 3 tables already created in mysql server which I’ve extracted.



Identify two or more stories from the sales data sheet that can be overlapped using joins and groups.

1) Sum of profit by Sub_Category and Country: This will help in understanding which product categories are performing well in each Country
2) Sum of revenue by Sub_Category. This will help us understand which subcategories are performing well overall.
3) Sum of order quantity ordered for each product: This helps us understand which product is doing well and which is not.
4) Customer Demographics by Country (Age and Gender): This will give insights into the customer base in each country, segmented by age and gender.

Split the sales data into two or more tables demonstrating the different stories

I have already created these tables and they are available through mysql server as well.

Sales Table

This table will include details about each sales transaction.

- **Columns**: Date, Country, Product, Order_Quantity, Unit_Cost, Unit_Price, Profit, Cost, Revenue.

Products Table

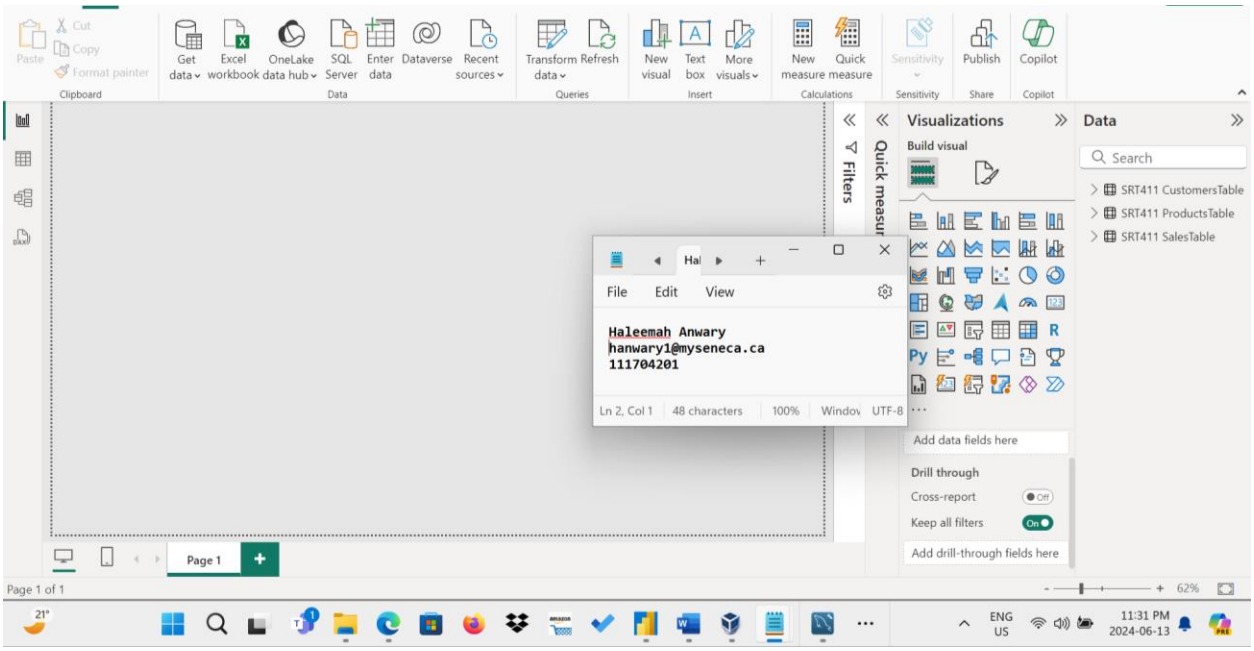
This table will contain information specific to the products

- **Columns:** Product, Product_Category, Sub_Category.

Customers Table

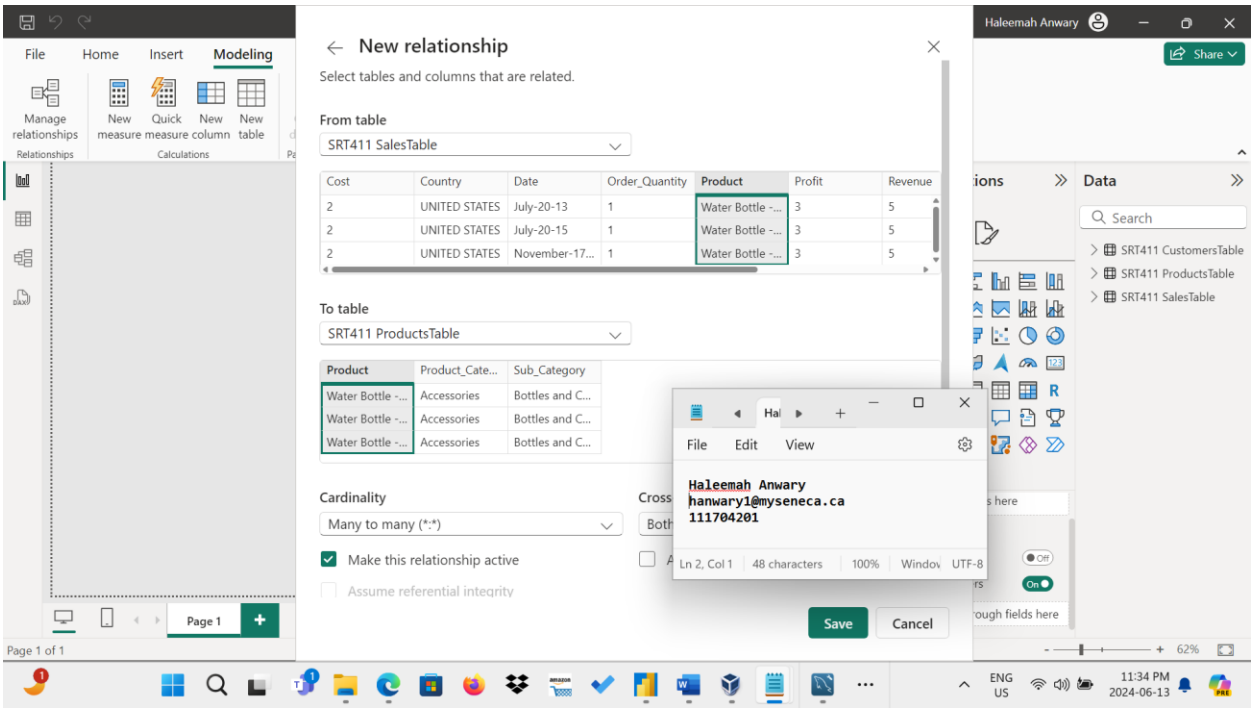
This table will capture customer demographics.

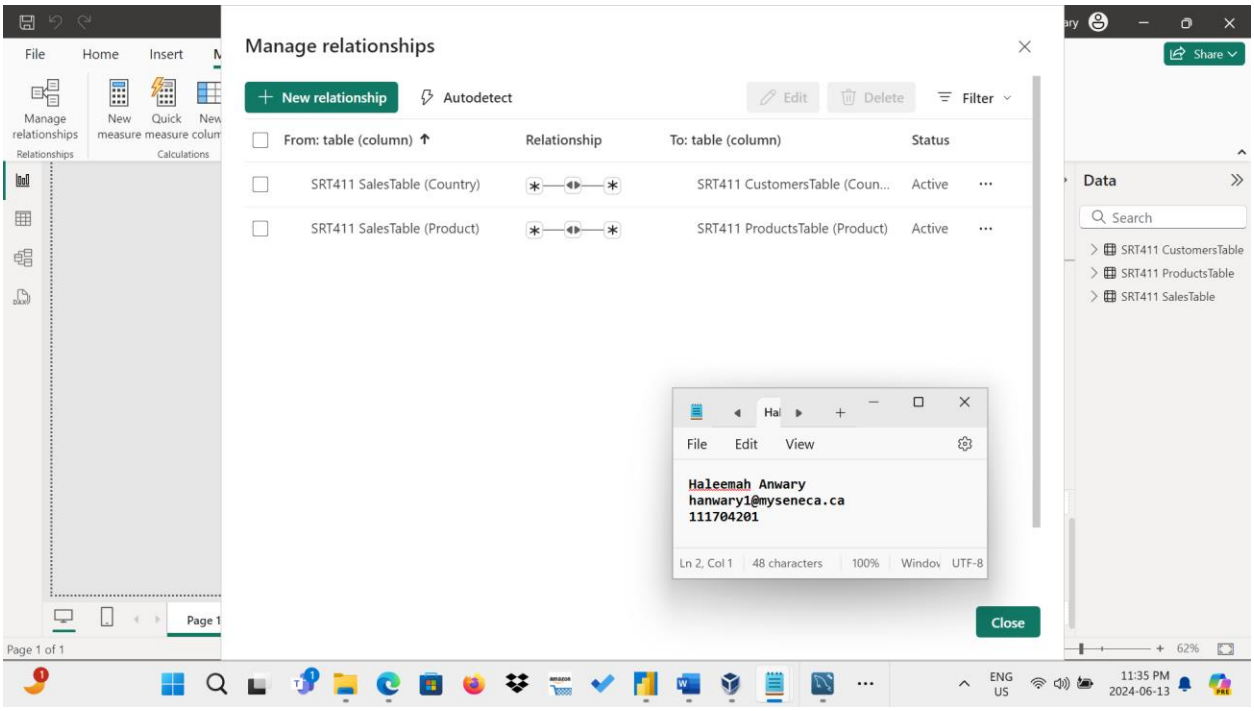
- **Columns**: Country, Customer_Age, Age_Group, Customer_Gender.



Create Relationships

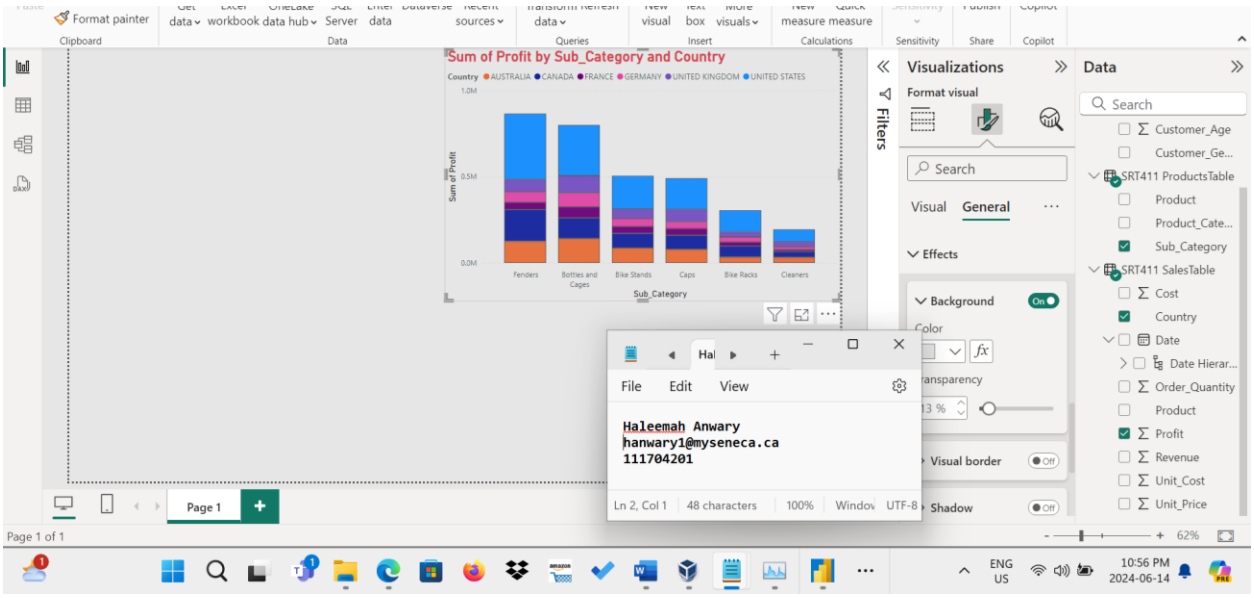
- **Sales** and **Products** via **Product**
- **Sales** and **Customers** via **Country**



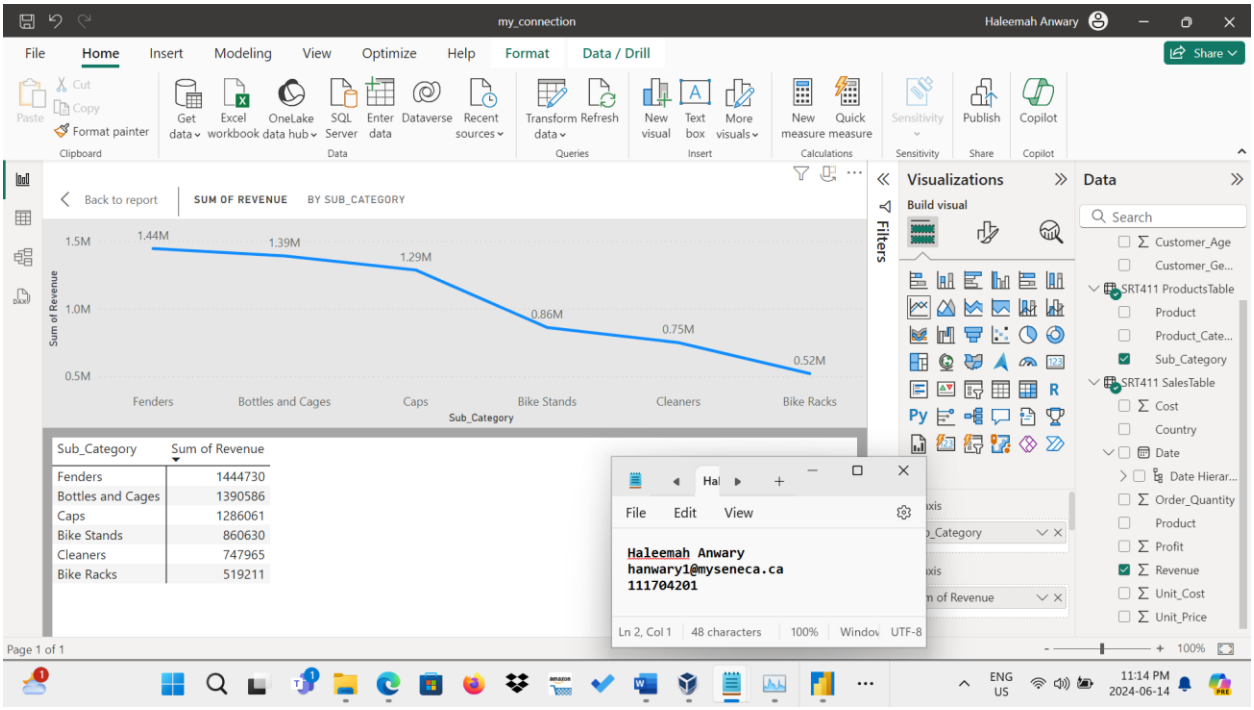


Create a data model using the different tables imported and explore data in Power BI. Explain and explore data and document the exploration.

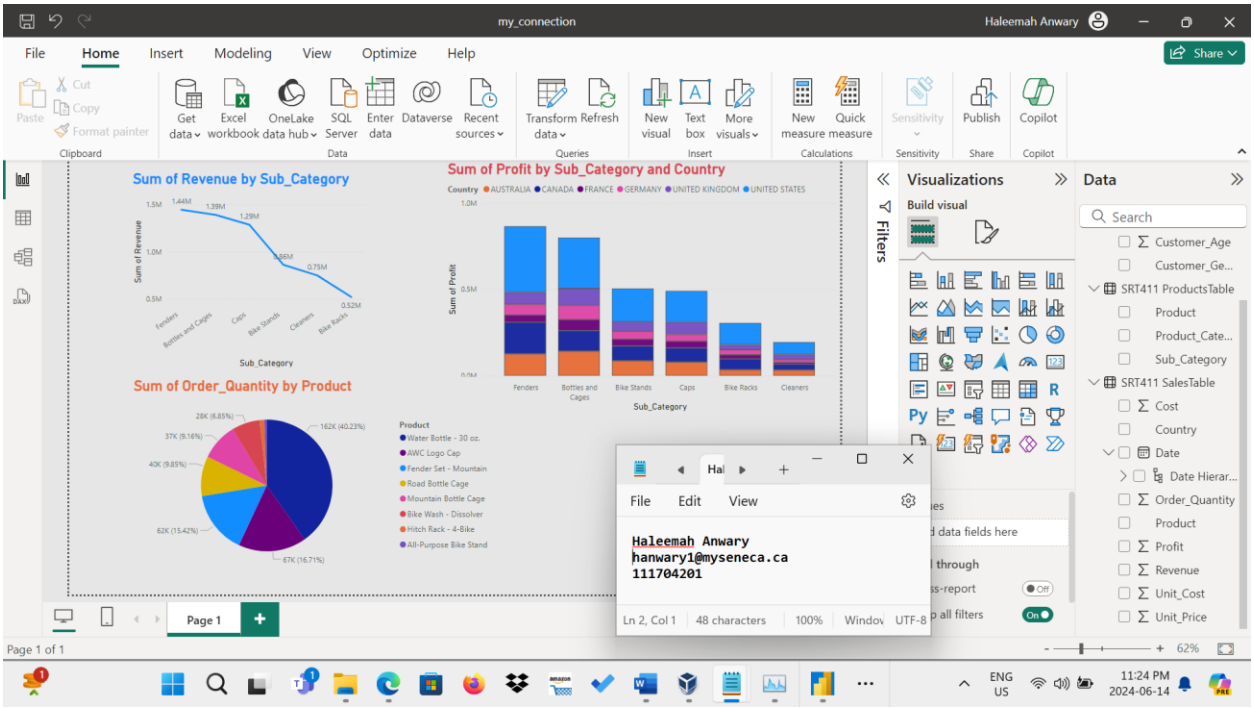
In this visual, I wanted to show the sum of profit by sub_category and Country. I used the visualization panel to apply a stacked column chart and turned on the titles by going to format visuals panel and different colors for each bar. By going into 'General' I edited the title color and font and made it bold to make it easy to read. Lastly, I changed the background color to be grey.



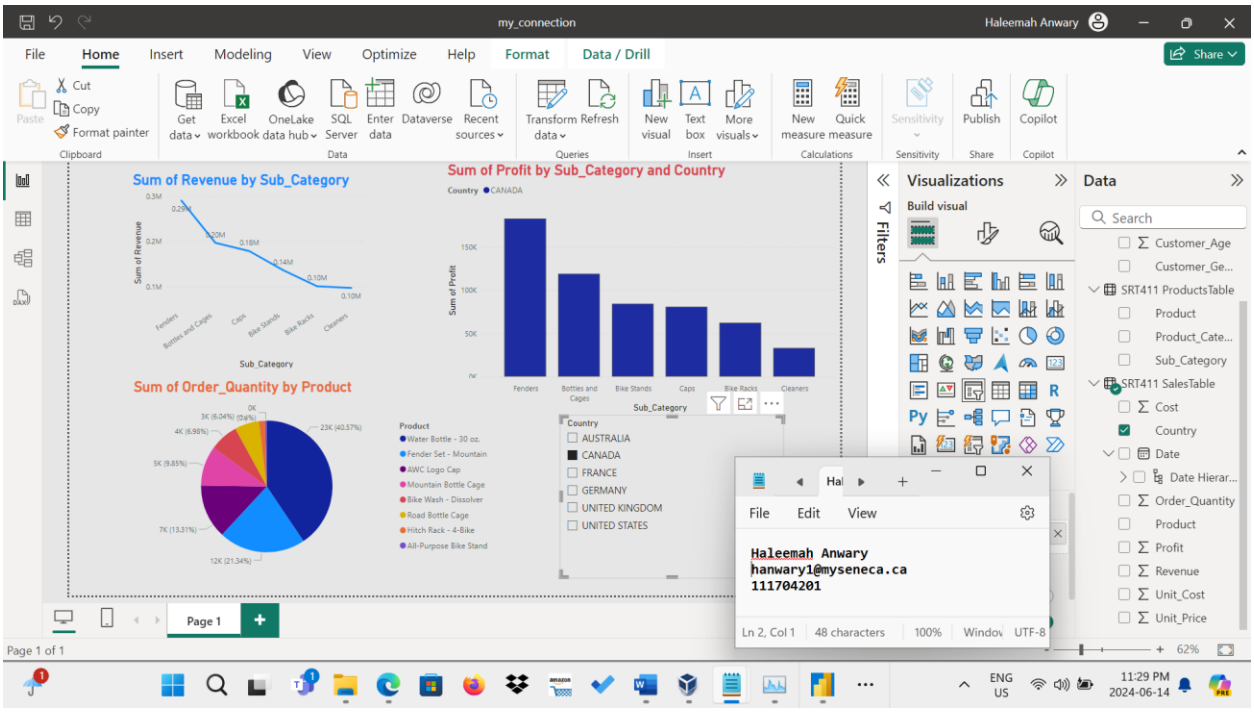
In this visual, I added a line chart to show the relationship between the revenue generated by each subcategory. The same as above chart, I added a background color, data labels, and edited the title.



In this chart, I am showing the total order quantity ordered for each product. To visualize this, I used a pie chart.

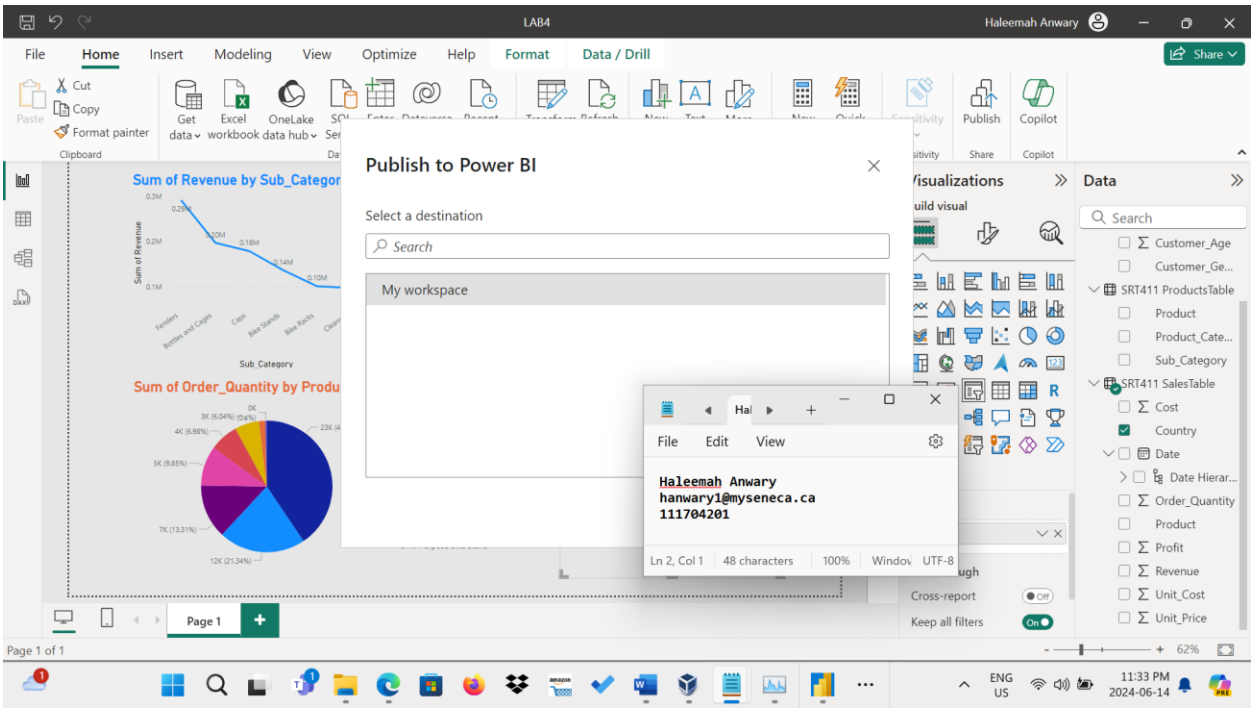


For the last visual, I used a slicer to filter the visuals by Country. I did this by dragging the country field and from visualizations, I set it as a slicer.

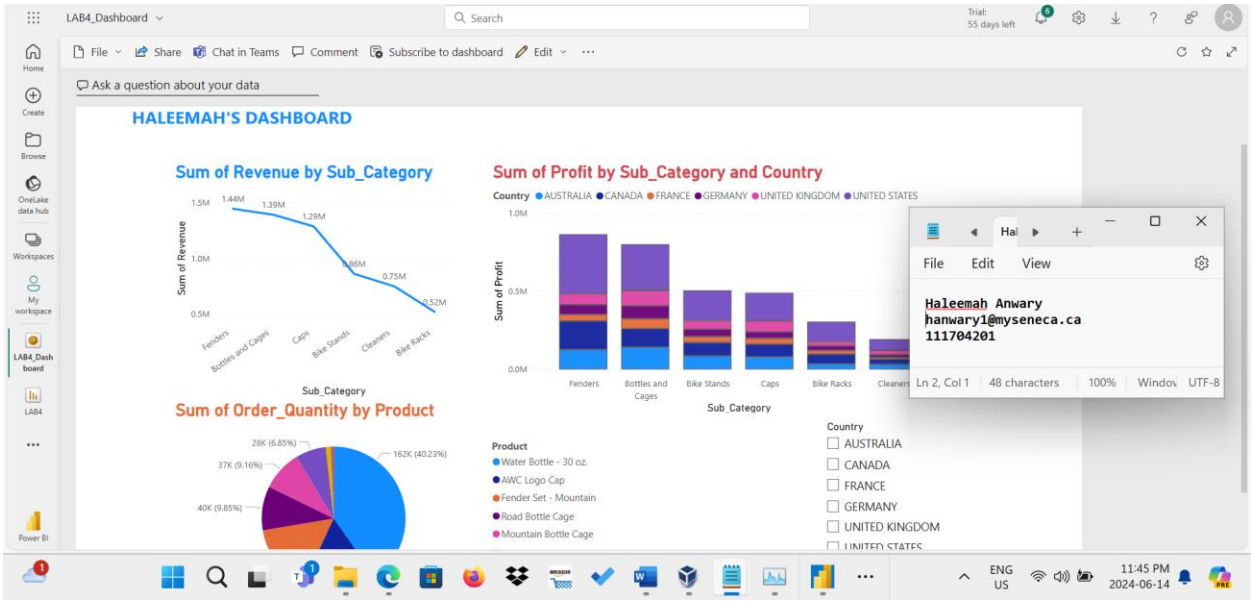


Using the Data Model and Built in the Visualisations Tools create a dashboard

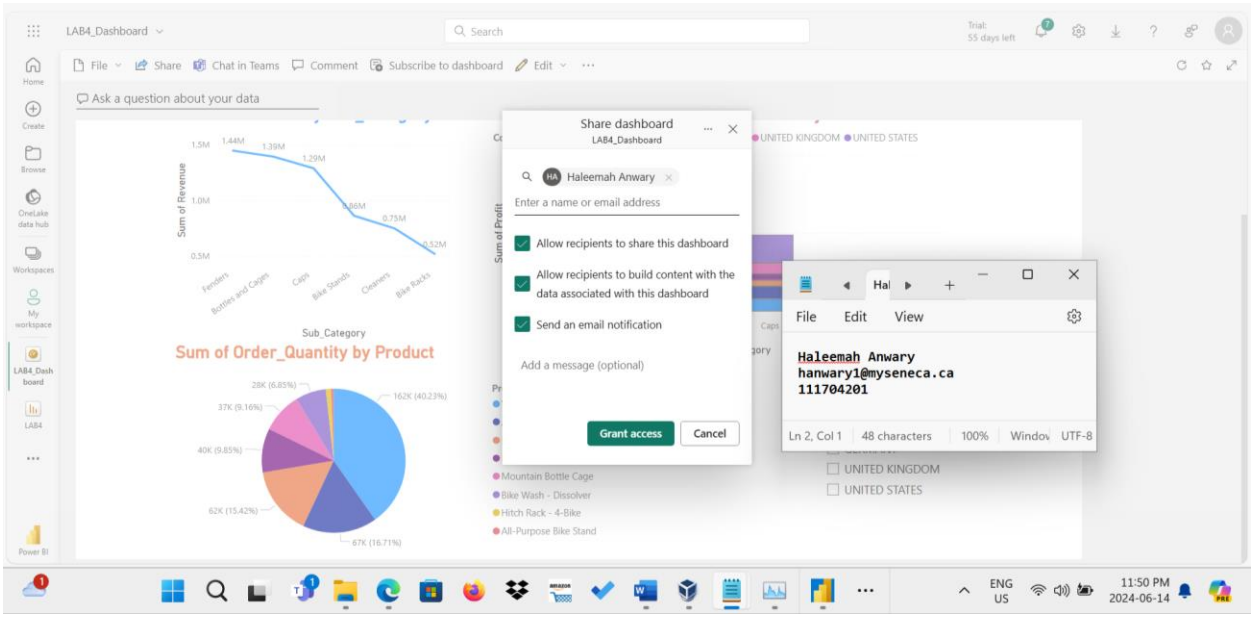
Publishing the report to ‘MYWORKSPACE’ on Power Bi



Created a dashboard and edited the visuals a bit further



Setup and Share Report to your Seneca Email



Learning Outcome on Power BI (Task 05)

Import Data

Power BI simplifies data importation from various sources like databases, cloud services, and files. Its "Get Data" feature guides users through connecting to and importing data seamlessly.

Transform Data

The Power Query Editor in Power BI allows users to clean and transform data using an intuitive interface. This includes operations like filtering, data type conversion, and merging datasets.

Model Data

Users can define relationships and calculate columns in Power BI using DAX (Data Analysis Expressions). This enhances the data model, allowing for more complex analyses and insights.

Create Visualizations

Power BI offers a wide range of visualization options, enabling users to create dynamic charts, graphs, and maps easily. It also supports custom visuals for tailored analytical needs.

Manage Dashboards

Dashboards in Power BI aggregate various visualizations into a single view, offering real-time insights. Users can customize dashboards with alerts and automatic updates.

Share and Collaborate

Power BI facilitates sharing and collaboration through features like:

- **Publishing and accessing reports** on Power BI Service for team viewing.
- **Collaborative workspaces** for joint development of reports and dashboards.
- **Integrated sharing** via email or applications like Microsoft Teams.

These functionalities make Power BI an effective tool for turning data into actionable intelligence and enhancing collaborative decision-making across an organization.

Learning Outcome on Tools and Analytics (Task 06)

Excel, SQL, and Power BI are pivotal in the data analytics landscape, each serving a unique role. Excel excels at handling and manipulating small datasets, perfect for initial data cleaning and exploration. SQL is crucial for managing larger datasets in relational databases, efficiently handling complex queries and data operations. Power BI leverages the strengths of both Excel and SQL to provide advanced visualization and business intelligence. It transforms complex data into interactive, accessible reports and dashboards that aid in making informed decisions. Together, these tools create a comprehensive data analytics workflow, allowing users to manage data from entry to decision-making effectively.

Learning Outcome In Data Driven Security (Task 07)

In data-driven security, mastering tools like Excel, SQL, and Power BI enables professionals to bolster cybersecurity measures effectively. Here are three concise use cases demonstrating their impact:

1. **Incident Analysis with SQL:**
    - **Use Case:** SQL is invaluable for quickly querying security logs to analyze incidents, helping pinpoint vulnerabilities and understand breach extents.
    - **Benefits:** Fast, precise query capabilities are crucial for effective incident response and mitigating security threats.
2. **Threat Detection with Power BI:**
    - **Use Case:** Power BI's real-time dashboards integrate data from multiple sources to highlight unusual activities, aiding in early threat detection.
    - **Benefits:** Its intuitive visual dashboards allow security analysts to easily spot and monitor potential threats.
3. **Compliance Reporting with Excel:**
    - **Use Case:** Excel is used extensively for risk assessments and compliance reporting, analyzing historical data to identify risks and ensure regulatory compliance.
    - **Benefits:** Excel's flexibility and familiarity make it ideal for detailed analyses and generating compliance reports efficiently.